

Necessary but Not Transplantable: Deep-Noise Robustness of Mamba-Style Blocks Is a Component-Interaction Effect

Yi Wang*

College of Engineering, Lishui University, Lishui 323000, Zhejiang, China

Abstract

The robustness of Mamba-style selective state-space models (SSMs) is widely attributed to their defining feature: input-dependent selectivity. We falsify this attribution through a three-step preregistered ablation chain on a controlled distribution-shift $\times$ noise testbed (bearing vibration signals under operating-condition drift). First, freezing selectivity—replacing input-dependent Δ , B , C with learned constants while keeping the block otherwise intact—costs only 2–5 pp under deep noise, yet the frozen block still outperforms a parameter-matched time-invariant S4D by 14–18 pp. Second, component-wise ablation localises the necessary ingredient: removing the multiplicative SiLU gate collapses performance below plain S4D. Third, however, grafting the gate onto S4D recovers merely 17% of the robustness gap—the gate is necessary within its native block but not transplantable. We conclude that deep-noise robustness is a component-interaction effect of the block composition, not a property of any single mechanism. Per-class analysis further shows the effect concentrates on the hard minority class, whose recall under noise is governed by gate presence. Our results caution against crediting input-selectivity for the matched-noise robustness of Mamba-style blocks, and against expecting any single transplanted component to confer that robustness on another backbone under a matched training budget.

Keywords: state-space models, Mamba, input selectivity, gating, robustness, ablation, distribution shift, component interaction

*Corresponding author

Email address: `jefflsxy@lsu.edu.cn` (Yi Wang)

1. Introduction

Selective state-space models (SSMs) of the Mamba family have rapidly become default sequence backbones, and a common narrative has formed around *why* they work: input-dependent selectivity—the ability to modulate state dynamics per token—is credited with filtering noise, adapting to distribution shift, and outperforming time-invariant predecessors such as S4D [1]. [2, 3] This belief motivates a growing family of “selective” architectures, yet it sits in unresolved tension with recent evidence: the first systematic S4D-versus-Mamba comparison on time-series classification finds the *simpler, time-invariant* model consistently superior [4], and theoretical analyses identify expressivity regimes where Mamba is provably weaker [5]. Which account is right—and, more importantly, *what exactly should be credited* when a Mamba-style block does prove robust? This paper answers the second question with a controlled, preregistered falsification chain, using bearing vibration diagnosis under operating-condition drift and matched noise as a testbed that makes the robustness gap large (up to 18 pp), reproducible (fingerprint-verified across 865 preregistered cells), and dissectible.

Our approach is deliberately *attributional* rather than *methodological*: we propose no new architecture and claim no state-of-the-art. Instead we take a regime in which a Mamba-style block exhibits a large, stable robustness advantage over a time-invariant SSM, and ask which ingredient earns it. The chain has three preregistered steps, each with adjudication thresholds fixed before the corresponding grid was run:

1. **Falsify selectivity (necessity test).** We construct a *frozen-selectivity* arm: the input-dependent projections producing Δ , B , C are replaced by learned constants, with every other component of the block—the convolutional stem, the multiplicative SiLU gate, the state parameterisation, normalisation, and projections—left byte-identical. If selectivity drives robustness, the frozen arm should collapse toward the time-invariant baseline.
2. **Localise the necessary component.** On the frozen base, we remove one component at a time (convolutional stem; SiLU gate; heavy-tailed state parameterisation) under a single-variable discipline verified key-by-key on the state dictionary, and adjudicate each arm against preregistered Δ -thresholds.

36 **3. Test sufficiency by constructive grafting.** The component iden-
 37 tified as necessary is transplanted onto plain S4D, and a preregistered
 38 recovery ratio decides whether “removal collapses” is matched by “ad-
 39 dition restores”.

40 The outcome is a pattern we summarise as *necessary but not trans-*
 41 *plantable*. Freezing selectivity costs only 2–5 pp in deep noise while the frozen
 42 block retains a 14–18 pp advantage over S4D—selectivity is not the mecha-
 43 nism. Removing the SiLU gate is catastrophic: the gateless block falls *below*
 44 plain S4D. Yet grafting the same gate onto S4D recovers only $R = 0.17$ of the
 45 gap, with essentially zero recovery at the deepest adjudicated noise level. The
 46 two results are symmetric and jointly decisive: the gate functions only in-
 47 side its native composition. Deep-noise robustness is therefore an interaction
 48 property of the block, not of any single part. Per-class analysis adds a final
 49 layer: the interaction effect concentrates on the hard minority class, whose
 50 recall trajectory under noise is governed by gate presence in both ablation
 51 directions.

52 **Contributions.** (i) A preregistered, fingerprint-verified falsification of
 53 the selectivity-robustness attribution for Mamba-style blocks under distri-
 54 bution drift and noise, reconciling the apparent conflict between selectivity-
 55 credit narratives and recent S4D-superiority findings [4]: both are consistent
 56 once robustness is credited to block composition rather than selectivity. (ii) A
 57 component-level attribution protocol—confound elimination (capacity, nor-
 58 malisation, auxiliary losses), frozen-selectivity construction at the module
 59 level without kernel modification, single-variable component removal, and
 60 constructive grafting—that upgrades “which architecture wins” comparisons
 61 into “which mechanism earns it” analyses. (iii) Evidence that the operative
 62 component (multiplicative gating) is necessary but not transplantable, with
 63 the effect concentrated on the hard class—a caution for the growing practice
 64 of transplanting Mamba components into other backbones expecting robust-
 65 ness gains. (iv) A reproducibility protocol combining evaluation-set finger-
 66 prints, preregistered adjudication thresholds, and dual-agent review, under
 67 which no post-hoc criterion change survived; we release the full adjudication
 68 trail.

2. Background and Related Work

2.1. Time-invariant and selective SSMs

Diagonal structured SSMs (S4D) [6] evolve a linear state with input-independent parameters (A , B , C , Δ), acting as a bank of learned linear filters. Mamba-style blocks [1] make Δ , B , C functions of the input token (selectivity), wrap the scan in a block containing a short causal convolutional stem, a multiplicative SiLU gate on a parallel branch, and residual/normalisation structure; the concrete block used throughout this paper follows the Mamba-3 formulation [7]. The narrative crediting selectivity for robustness draws on theory showing input-dependent dynamics can represent discontinuity-sensitive projections and counteract memory decay [8], and on [9]. We treat these as *hypotheses to be tested*, not established attributions: the theoretical results concern representability, which does not by itself determine which component earns empirical robustness in a trained block.

2.2. The unresolved comparison

Empirically the picture is contested. A systematic 59-dataset comparison finds S4D-based models superior to Mamba variants on clean time-series classification [4]; theoretical work identifies long-range regimes where Mamba is weaker than S4D [5]. Meanwhile, application literatures—including machine-condition monitoring—contain many Mamba-hybrid methods reporting noise robustness [10, 11, 12, 13, 14], but none, to our knowledge, isolates *which component* of the block is responsible under controlled conditions. Our clean-data results agree with the S4D-superiority finding (S4D matches or slightly exceeds the Mamba-style block on clean data); the disagreement in the literature dissolves once the drift $\times$ noise regime is examined separately and the credit is assigned at component level.

2.3. Testbed rationale

We use bearing vibration diagnosis under cross-speed operating-condition drift as the testbed, for three properties that matter to the attribution question rather than to the application: (a) the distribution shift is physically generated (rotation-speed change), not simulated; (b) fault signatures are sparse transients—precisely the discontinuous structures selectivity theory predicts selective SSMs should excel at [8], making it a *favourable* arena for the selectivity hypothesis; and (c) the robustness gap between the Mamba-style block and S4D is large (up to 20 pp), giving the attribution chain high

104 signal. The testbed is two-class (details in Section 3.1); the scope of all claims
 105 is bounded accordingly (Section 6).

106 3. Experimental Protocol

107 3.1. Testbed, arms, and budget

108 Vibration windows from the XJTU-SY run-to-failure dataset [15] are split
 109 by operating condition: models train on one speed condition and evaluate on
 110 another (cross-speed), with additive white Gaussian noise applied at matched
 111 SNR to training and evaluation ($\{\text{clean}, +10, +6, 0, -2, -6, -10 \text{ dB}\}$). The
 112 task is binary fault classification (inner-race vs. outer-race dominant classes;
 113 class imbalance makes IR the hard class, Section 4.4). All arms share an
 114 identical training budget (20 epochs, 5 seeds), identical splits, and identi-
 115 cal evaluation data verified by a byte-level fingerprint: every reported cell
 116 carries `eval.sha = 6c20b367...` (Appendix Appendix A), recomputed inde-
 117 pendently during adjudication. In total the study comprises 865 cells across
 118 five preregistered grids (baseline grid 475; frozen-selectivity 105; component
 119 ablation 175; graft 35; per-class 75).

120 The principal arms are summarised in Table 1.

121 The frozen-selectivity construction requires no kernel modification: the
 122 selective-scan interface accepts Δ , B , C tensors regardless of their origin,
 123 so input-dependent projections are replaced at the module level by learned
 124 constants broadcast over time; the removed projections are deleted from the
 125 module so no parameter or gradient can leak. Single-variable discipline is
 126 enforced by key-level state-dictionary diffs: for every ablation arm, all keys
 127 outside the targeted component are verified byte-identical, and every differing
 128 key is mapped to an explicit entry in an architecture map.

129 3.2. Confound elimination

130 Before component attribution, three confounds were eliminated on the
 131 same grid. **Capacity:** S4D-wide ($1.75\times$ parameters) tracks S4D pointwise
 132 (mean -0.5 pp)—parameters do not buy robustness at this scale; note also
 133 the gateless arm (79,642 params) has *fewer* parameters than S4D yet the
 134 gate question is decided in the opposite direction (Section 4.2). **Normal-**
 135 **isation:** both SSM arms use LayerNorm and contain no BatchNorm; the
 136 arm that degrades under noise is the one *without* any batch statistics, ruling
 137 out the BatchNorm-failure narrative as the mechanism. **Auxiliary physics**
 138 **loss:** an exploratory kinematics-alignment regulariser carried by the original

Table 1: Principal experimental arms and parameter counts.

Arm	Description	Params
BM3	Mamba-style block stack (conv stem $k=7 \rightarrow$ selective scan $\rightarrow$ SiLU gate $\rightarrow$ projection), with and without an auxiliary physics loss (Section 3.2)	177,938
BM3-frozen	BM3 with Δ , B , C replaced by learned constants (selectivity off); all other components unchanged	112,410
S4D	Parameter-matched diagonal time-invariant SSM (LayerNorm; no BatchNorm anywhere in either SSM arm)	100,162
S4D-wide	S4D widened to $1.75\times$ parameters (capacity control)	178,498
frozen-conv	Single-component removal on the frozen base: conv stem (Section 4.2)	112,026
frozen-gate	Single-component removal on the frozen base: SiLU gate (Section 4.2)	79,642
frozen-A	Single-component removal on the frozen base: heavy-tailed A parameterisation (Section 4.2)	112,410
S4D+gate	Plain S4D plus the SiLU gate branch only (graft, Section 4.3)	116,802
CNN/TCN baselines	Conventional baselines for context	—

139 model contributes -0.1 pp pointwise (curves coincide); all attribution arms
 140 are therefore built on the pure cross-entropy variant.

141 3.3. Preregistration and adjudication

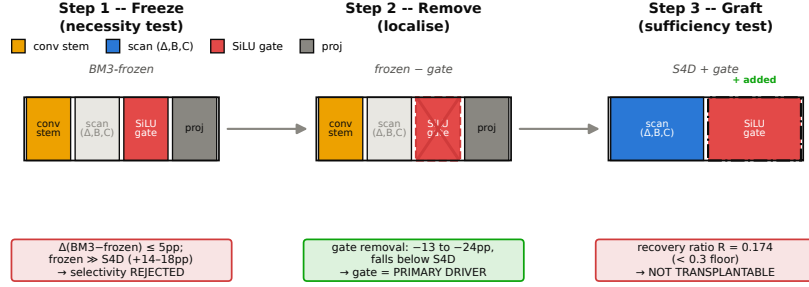
142 Every adjudicated question in this study had numeric decision thresh-
 143 olds fixed and written to disk before the corresponding grid was launched
 144 (file timestamps precede run-directory timestamps and are released, Ap-
 145 pendix [Appendix A](#)). Adjudication was mechanical: an analysis pass applies
 146 the thresholds and emits a decision artifact; an independent reviewer pass
 147 re-computes aggregates, verifies fingerprints row-by-row, and must cite file
 148 paths and modification times for every check. The preregistered thresholds
 149 were binding in practice: two attempted post-hoc criterion substitutions and
 150 one stale-artifact false verification were caught and re-adjudicated under the
 151 original rules — this is the one place where preregistration actually bit. Ap-
 152 pendix [Appendix C](#) details the incidents, the automated executor/reviewer
 153 implementation, and the human adjudication gates; all scientific decisions —
 154 thresholds, rulings, and interpretations — were made by the authors and are
 155 timestamped. Statistical tests (paired, per-condition) are reported through-
 156 out but were never adjudication criteria. Process-level reproducibility has
 157 a measurable floor: identical configurations re-run in separate processes dif-
 158 fer by ~ 0.7 pp macro-F1 (non-deterministic kernel reductions), an order of
 159 magnitude below seed-level variability (2–9 pp); all within-grid comparisons
 160 share batch and process conditions ([Appendix B](#)).

161 4. Results

162 Throughout, “deep noise” denotes the preregistered adjudication band
 163 $\{0, -2, -6 \text{ dB}\}$; -10 dB is reported but excluded from adjudication (the time-
 164 invariant baseline is at chance there). All values are macro-F1 (%), mean
 165 over 5 seeds. [Figure 1](#) summarises the three-step falsification chain that
 166 structures this section.

167 4.1. Selectivity is not the mechanism

168 Freezing selectivity costs at most 5.1 pp anywhere in the adjudication
 169 band, while the frozen block retains a 13.8–17.7 pp advantage over S4D ([Ta-
 170 ble 2](#), [Figure 2](#)). Under the preregistered rule (attribution confirmed only if
 171 $\Delta(\text{BM3} - \text{frozen}) \geq 10$ pp in deep noise with the frozen arm collapsing toward
 172 S4D), the selectivity attribution is **rejected**. On clean data S4D matches



Deep-noise robustness is a component-interaction effect of the block composition, not of any single mechanism.

Figure 1: The three-step preregistered falsification chain: freeze selectivity (Section 4.1), remove one component at a time on the frozen base to localise the necessary ingredient (Section 4.2), and graft that ingredient onto S4D to test sufficiency (Section 4.3).

Table 2: BM3 vs. BM3-frozen vs. S4D, macro-F1 (%) by SNR. $\Delta(\text{BM3-frozen})$ and $\Delta(\text{frozen-S4D})$ in percentage points.

Condition	BM3	BM3-frozen	S4D	$\Delta(\text{BM3-frozen})$	$\Delta(\text{frozen-S4D})$
clean	78.9	78.9	81.0	-0.0	-2.1
0 dB	95.1	89.9	76.1	+5.1	+13.8
-2 dB	95.4	92.1	75.2	+3.3	+17.0
-6 dB	88.5	86.3	68.6	+2.2	+17.7
-10 dB*	68.1	64.3	52.3	+3.8	+12.0

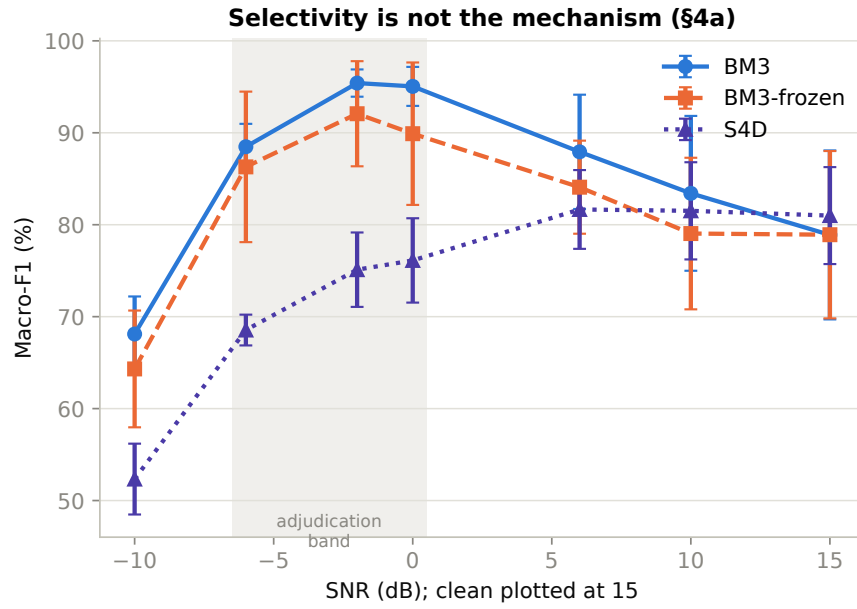


Figure 2: Macro-F1 vs. SNR for BM3, BM3-frozen, and S4D across the full noise grid. The frozen arm tracks BM3 closely and stays well above S4D throughout the adjudication band, falsifying the selectivity attribution.

173 or slightly exceeds both Mamba-style arms—consistent with the systematic
 174 clean-TSC comparison [4]—so the literature’s conflict is a regime effect, and
 175 within the noisy regime the credit must lie elsewhere in the block.

176 4.2. The gate is the necessary component—and the only one

Table 3: Single-component removal on the frozen base, $\Delta(\text{frozen} - \text{arm})$ in percentage points.

Component removed	0 dB	−2 dB	−6 dB	Adjudication
SiLU gate	+13.1	+21.1	+23.5	PRIMARY (all ≥ 8 pp; arm falls to/below S4D)
conv stem ($k=7$)	−6.6	−2.7	+7.9	outside preregistered bins (regime trade-off)
heavy-tailed $A \rightarrow$ S4D-style A	−2.4	−2.2	−4.2	outside bins (removal beneficial)

177 The gate result is unambiguous and monotone in noise depth; critically,
 178 the gateless arm does not merely lose the Mamba advantage—it falls **be-**
 179 **low plain S4D** at −2 dB (71.7 vs. 75.2) and −6 dB (62.5 vs. 68.6). A
 180 conv-plus-constant-SSM composition without gating is worse than the sim-
 181 pler time-invariant model, a first hint that components interact rather than
 182 add. The other two components tell regime stories rather than robustness
 183 stories: the convolutional stem pays +7.9 pp at −6 dB but *costs* 21.0 pp on
 184 clean cross-speed data (removal yields 98.8 vs. 77.8, with seed-std collaps-
 185 ing from 8.1 to 0.6 pp)—consistent either with speed-specific local temporal
 186 patterns mistransferring across conditions, or with optimisation friction un-
 187 der the matched budget; we report both readings. The heavy-tailed state
 188 parameterisation is mildly harmful throughout (its S4D-style replacement is
 189 the best deep-noise cell in the whole component grid, 90.1 at −6 dB).

190 4.3. Necessary but not sufficient: the graft fails

191 Preregistered recovery ratio $R = \text{mean over the band of } (\text{S4D} + \text{gate} -$
 192 $\text{S4D}) / (\text{frozen} - \text{S4D}) = \mathbf{0.174}$, far below the 0.7 confirmation threshold; at
 193 the deepest adjudicated level the graft recovers essentially nothing. Together
 194 with Section 4.2 this closes a symmetric argument: *removal collapses below*

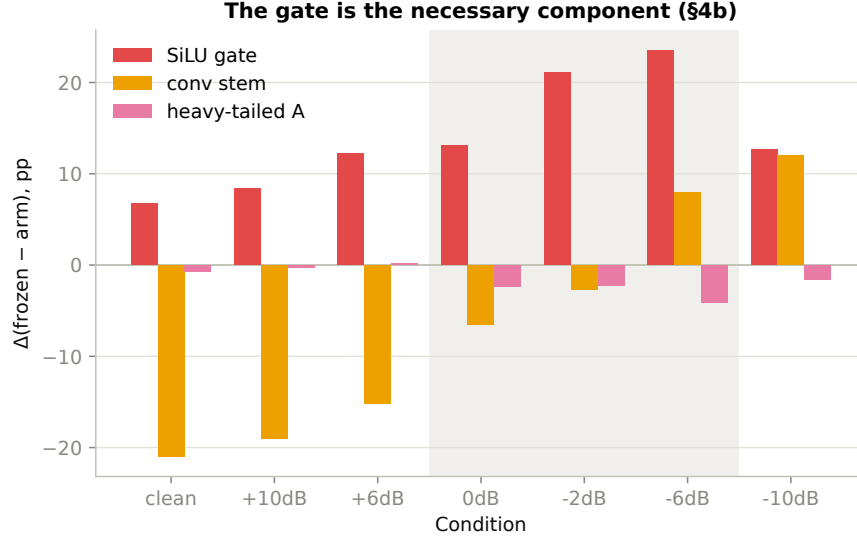


Figure 3: Component-removal Δ (percentage points, frozen – arm) by SNR for the three single-component ablations on the frozen base. Only gate removal is monotone, large, and one-signed across the adjudication band.

Table 4: Grafting the SiLU gate branch onto plain S4D (S4D unchanged key-by-key; only the gate branch added).

Condition	S4D+gate	S4D	BM3-frozen	Recovery ratio
0 dB	79.8	76.2	90.6	0.249
−2 dB	79.4	75.2	92.8	0.239
−6 dB	69.2	68.6	86.0	0.035

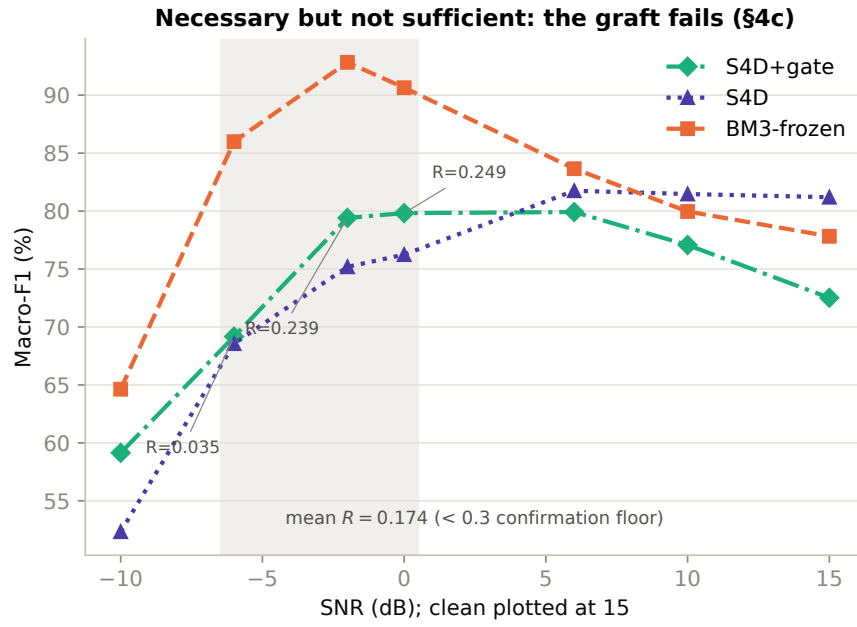


Figure 4: Graft outcome: S4D+gate, plain S4D, and BM3-frozen across the adjudication band, with the per-condition recovery ratio $R = (S4D+gate - S4D) / (frozen - S4D)$ annotated. Recovery collapses to near zero at the deepest adjudicated noise level.

195 *the baseline; addition barely moves it.* If the gate were a self-contained ro-
 196 bustness mechanism, both directions would move by comparable amounts.
 197 Instead the gate functions only inside its native composition. Two readings
 198 are compatible with the graft failure and we report both: genuine interac-
 199 tion dependence (the gate needs the conv stem / constant-scan context it
 200 co-trained with), and transplant friction (the grafted branch may be harder
 201 to optimise in a foreign host under the matched budget; training converges
 202 normally, but convergence is not robustness). Under either reading, the prac-
 203 tical conclusion stands: single-component transplants should not be expected
 204 to confer the donor block’s robustness.

205 4.4. The interaction effect concentrates on the hard class

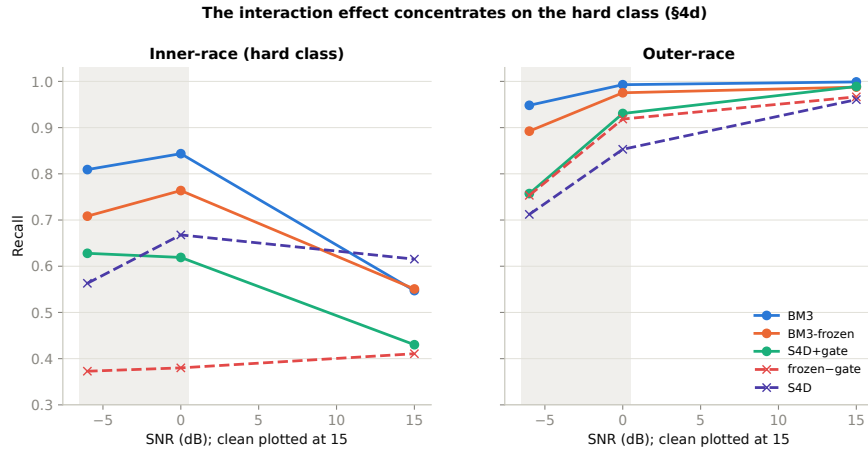


Figure 5: Per-class recall (inner-race, IR, and outer-race, OR) as a function of noise depth, split by gate presence. Gated arms lift IR recall under matched noise; gateless arms let it deteriorate.

206 Per-class analysis ($5 \text{ arms} \times \{\text{clean}, 0, -6 \text{ dB}\} \times 5 \text{ seeds}$, per-sample
 207 predictions dumped and re-verified against reported metrics cell-by-cell): the
 208 inner-race class is hard from the outset (clean recall 0.41–0.62 vs. 0.96–1.00
 209 for outer-race). Its trajectory under noise is governed by gate presence: in
 210 all three gated arms IR improves under matched noise, while in both gateless
 211 arms it deteriorates (Figure 5). Both ablation directions hit IR hardest—gate
 212 removal costs IR 2.6–3.8 $\times$ more F1 than OR at every level, and the graft’s
 213 noise-time gains also concentrate on IR (−6 dB: +6.8 pp IR vs. +4.2 pp OR).
 214 Because macro-F1 averages the two classes, the arm-level robustness gaps of

Sections 4.1–4.3 are carried substantially by the hard class. This refines the headline claim: the component-interaction effect operates through the low-margin class—multiplicative gating appears to protect precisely the class with the least separability margin—and it is honest to state that the effect is class-concentrated rather than uniform.

5. Discussion

What should be credited? The three-step chain licenses a specific, narrow attribution: within this block, on this testbed, deep-noise robustness is earned by the composition—gate $\times$ (stem, scan, parameterisation)—with the gate as the necessary but non-transplantable ingredient, and with selectivity, capacity, normalisation and the physics loss all eliminated as explanations. The apparent contradiction between “selectivity makes Mamba robust” narratives and S4D-superiority findings [4] dissolves under this credit assignment: both literatures are right in their regimes, and neither regime is about selectivity.

Implications for backbone selection and for transplant practice. For practitioners choosing backbones for noisy, drifting signal regimes: what a Mamba-style block buys is its composition, and its cost structure should be judged accordingly (the gate cannot be cherry-picked into a cheaper host; our graft numbers quantify the failure). For the growing practice of porting Mamba components into CNNs, Transformers, or hybrid diagnosis models with an expectation of robustness gains, Section 4.3 is a directly usable negative result. A falsifiable engineering prediction follows: a recomposed block guided by the component map (no conv stem, gate retained, S4D-style A) should dominate the original on this testbed; we deliberately leave this untested here, as post-hoc recomposition after seeing the map would not carry the evidential status of the preregistered chain.

Why might gating be composition-bound? We refrain from a mechanistic claim beyond the data, but note three compatible observations. First, the gate’s benefit concentrates on the low-margin class (Section 4.4), and matched-noise training lifts that class *only when the gate is present*—suggesting the gate is the component through which matched-noise training is converted into hard-class margin, i.e., an exploiter of noise-aware training rather than a test-time denoiser (this reading is bounded by the matched-SNR scope, Section 6). Second, its role appears closer to adaptive per-sample feature suppression than to generic denoising. Third, the gateless composi-

tion falling *below* S4D indicates the other components are calibrated to a gated regime (e.g., the stem may pass features that are only safe when the gate can veto them). All are testable in future work via pairwise interaction grids and mismatched-noise protocols.

6. Boundaries and Limitations

We state the scope plainly.

1. **Testbed scope.** All claims are bounded to cross-speed bearing vibration, binary classification, GPU execution (no edge/real-time claims); multi-class and cross-dataset replication is future work.
2. **Attribution level.** The credit assigned to “the rest of the composition” is by elimination within the enumerated component set; pairwise interaction terms are not decomposed.
3. **The transplant conclusion is budget-conditional.** All arms share a matched 20-epoch budget; the grafted gate trains normally yet recovers little, and convergence is not robustness (Section 4.3)—but we cannot exclude that a longer or graft-tuned budget would recover more. “Not transplantable” therefore means *not transplantable under a matched training budget*, and the headline should be read with this condition attached.
4. **Matched-SNR scope.** Training and evaluation share the same injected SNR throughout (Section 3.1); “robustness” in this paper is matched-noise robustness under noise-aware training, not clean-train/noisy-test generalisation. The gate’s benefit may operate by exploiting matched-noise training (consistent with Section 4.4) rather than by test-time denoising; attribution under mismatched-noise protocols is untested and may differ.
5. **Graft dual reading.** Interaction dependence vs. transplant friction (Section 4.3), the latter now subsumed under the budget condition of item 3.
6. **Noise-regime coverage.** Under strong coloured noise at -10 dB a TCN baseline slightly overtakes the Mamba-style block, and -10 dB is near-chance for the time-invariant baseline and excluded from adjudication; the demonstrated advantage regime is matched, Gaussian-dominated noise.

285 **7. Conv-stem dual reading.** Its clean-data cost admits both representation-
286 mistransfer and optimisation-difficulty readings, reported in place (Sec-
287 tion 4.2).

288 Process-level non-determinism (~ 0.7 pp, an order below seed variance)
289 and the full fingerprint/adjudication trail are documented in the reproducibil-
290 ity appendix (Appendices [Appendix A](#)–[Appendix B](#)).

291 **7. Conclusion**

292 Input-dependent selectivity does not earn the deep-noise robustness of
293 Mamba-style blocks on a controlled drift $\times$ noise testbed: freezing it costs
294 2–5 pp while 14–18 pp of advantage over a time-invariant SSM remains. The
295 multiplicative gate is the necessary component—its removal drops the block
296 below plain S4D—yet it is not transplantable: grafted onto S4D it recovers
297 17% of the gap, and essentially nothing at the deepest adjudicated noise
298 level. Robustness here is a property of the composition, concentrated on the
299 hard class, and earned by interaction rather than by any single mechanism.
300 We offer the three-step chain—freeze, remove, graft—as a reusable protocol
301 for turning architecture comparisons into component attributions, and our
302 preregistered adjudication trail as evidence that such attributions can be
303 made resistant to post-hoc rationalisation.

304 **CRedit authorship contribution statement**

305 Yi Wang: Conceptualization, Methodology, Software, Validation, Formal
306 analysis, Investigation, Data curation, Writing – original draft, Writing –
307 review & editing, Visualization, Supervision, Project administration.

308 **Declaration of competing interest**

309 The authors declare that they have no known competing financial inter-
310 ests or personal relationships that could have appeared to influence the work
311 reported in this paper.

312 Declaration of Generative AI and AI-assisted technologies in the 313 writing process

314 During the preparation of this work the authors used LLM-based agents
315 to execute and cross-check preregistered analysis and adjudication passes as
316 described in Appendix C, and to assist with manuscript drafting and figure
317 generation. After using these tools, the authors reviewed and edited the con-
318 tent as needed and take full responsibility for the content of the published
319 article. All numeric thresholds, adjudication rulings, and scientific interpre-
320 tations reported in this paper were fixed and made by the human authors,
321 as detailed in Appendix C.

322 Data availability

323 The code, preregistration artifacts, and adjudication trail for this study
324 are available at <https://github.com/jeff-1980/bm3-jizi>.

325 Appendix A. Fairness protocol

326 *Appendix A.1. Evaluation-set fingerprint*

327 Every reported cell in this study is tagged with `eval_sha256`, a SHA-
328 256 hash of the raw held-out test-label byte array, computed once per run
329 and asserted identical across every arm/condition/seed cell within a grid. It
330 fingerprints *that the same held-out label vector was scored every time*, not
331 model weights or predictions. The full value shared by all 865 cells across all
332 five grids is:

333 `6c20b367522ce5db59720120f8a7cd12792c7e070c6ae84bd1985511998727de`

334 This value is recomputed independently at adjudication time (not merely
335 copied from the run log) and cross-checked row-by-row against every grid’s
336 fairness report before any comparison across arms is treated as valid.

337 *Appendix A.2. Per-arm parameter counts*

338 Table A.5 lists every arm’s parameter count, each obtained directly via
339 `sum(p.numel() for p in model.parameters())` on the instantiated model
340 (not estimated), cross-verified against an independently maintained architec-
341 ture map.

Table A.5: Per-arm parameter counts (verified against `arch_map.md` and harness source).

Arm	Parameters
BM3 (<code>bm3_kin</code> / <code>bm3_nokin</code>)	177,938
BM3-frozen (<code>bm3_frozen</code>)	112,410
S4D (<code>s4d</code>)	100,162
S4D-wide (<code>s4d_wide</code> , capacity control, $1.75\times$)	178,498
frozen-conv (<code>frozen_noconv</code>)	112,026
frozen-gate (<code>frozen_nogate</code>)	79,642
frozen-A (<code>frozen_As4d</code>)	112,410
S4D+gate (<code>s4d_plus_gate</code>)	116,802

342 *Appendix A.3. Five-batch preregistration timeline*

343 Table A.6 lists, for each of the five preregistered grids, the artifact that
 344 fixed its adjudication thresholds, that artifact’s creation/modification date,
 345 and the corresponding grid’s run-directory start time. In every case the
 346 prereg artifact predates the grid it governs.

347 Two honest caveats. First, the baseline batch predates the orchestration
 348 infrastructure used for the other four batches by five days; its prereg trail
 349 rests on the task card’s creation timestamp alone, not on the same automated
 350 executor/reviewer machinery documented in Appendix Appendix C, and we
 351 state this rather than implying a uniform process across all five batches.
 352 Second, for the component-ablation grid, two additional JSON files bearing
 353 “prereg” in their name exist on disk (`secondary_prereg.20260704-0708` and
 354 `secondary_deltapp_prereg.20260704-0719`) that *post-date* the grid’s start
 355 time; both are explicitly marked void in the corresponding decision artifact’s
 356 `void_preregs` field, and neither was used to adjudicate any result in this
 357 paper. They are the on-disk residue of the first post-hoc criterion substitution
 358 described in Appendix Appendix C, retained here (rather than deleted) as
 359 part of the released adjudication trail.

360 **Appendix B. Confusion matrices and process-level reproducibil-** 361 **ity floor**

362 *Appendix B.1. Full confusion-matrix set*

363 Table B.7 gives the complete confusion-matrix set underlying the per-
 364 class analysis of Section 4.4: 5 arms $\times$ 3 conditions (clean, 0 dB, -6 dB),

Table A.6: Preregistration predates data collection for all five grids.

Grid (cells)	Canonical prereg artifact	Prereg date	Grid start
Baseline (475)	Task-tracking-system card (<code>created: 2026-06-27</code> , mtime 15:02)	2026-06-27	2026-06-27 17:31
Frozen-selectivity (105)	<code>frozensel_prereg.json</code> (mtime 18:26:54)	2026-07-02	2026-07-02 18:27
Component ablation (175)	Task-tracking-system card (<code>created: 2026-07-03</code>), materialised in <code>prereg_provenance.md</code>	2026-07-03	2026-07-03 10:34
Graft (35)	<code>graft_prereg_provenance.md</code> (mtime 21:20:36)	2026-07-05	2026-07-05 22:41
Per-class (75)	Task-tracking-system card (<code>created: 2026-07-06</code>)	2026-07-06	2026-07-07 21:21

each summed over 5 seeds (support OR= 17,520, IR= 8,720 per cell). These counts were dumped per-sample and re-verified against the reported recall/precision/F1 metrics cell-by-cell before use.

Table B.7: Full confusion-matrix set (summed over 5 seeds; support OR=17,520, IR=8,720 per cell). Columns give true→predicted counts.

Arm	Condition	OR→OR	OR→IR	IR→OR	IR→IR
BM3	clean	17,500	20	3,948	4,772
	0 dB	17,395	125	1,364	7,356
	−6 dB	16,613	907	1,664	7,056
BM3-frozen	clean	17,303	217	3,917	4,803
	0 dB	17,089	431	2,060	6,660
	−6 dB	15,634	1,886	2,543	6,177
S4D+gate	clean	17,327	193	4,969	3,751
	0 dB	16,303	1,217	3,322	5,398
	−6 dB	13,267	4,253	3,244	5,476
frozen−gate	clean	16,935	585	5,139	3,581
	0 dB	16,092	1,428	5,406	3,314
	−6 dB	13,200	4,320	5,469	3,251

Arm	Condition	OR→OR	OR→IR	IR→OR	IR→IR
S4D	clean	16,828	692	3,354	5,366
	0 dB	14,945	2,575	2,897	5,823
	−6 dB	12,477	5,043	3,810	4,910

368 *Appendix B.2. Process-level reproducibility floor*

369 Because two independent process runs of an identical configuration are
370 not bitwise reproducible on this hardware (non-deterministic GPU kernel re-
371 ductions in the selective-scan and normalisation ops), we measured a process-
372 level noise floor directly: the same arm, condition, and seed (`bm3_frozen`,
373 clean, seed 0, 2 epochs) was run twice in separate processes with identical
374 evaluation fingerprints and identical parameter counts confirmed. The two
375 runs’ best macro-F1 differed by

$$|0.69025 - 0.69712| = 0.00687 \approx 0.687 \text{ pp},$$

376 which we report as the ~ 0.7 pp process floor cited in Section 3.3. This is
377 an order of magnitude below the seed-level variability observed within the
378 deep-noise band across the curve tables underlying Figures 2 and 4 (standard
379 deviations typically 2–9 pp, e.g. BM3-frozen 5.7–9.4 pp and S4D 1.7–4.5 pp at
380 $\{0, -2, -6\}$ dB). All within-grid comparisons in this paper share batch and
381 process conditions, so this floor bounds — but does not explain away — the
382 arm-level effects reported in Section 4.

383 **Appendix C. Adjudication incidents, automation disclosure, and** 384 **human adjudication gates**

385 This appendix gives the material Section 3.3 summarises in one sentence:
386 what the preregistration discipline actually caught, how the analysis/review
387 pipeline was implemented, and exactly where human judgement entered. We
388 report this in full rather than in summary because the growing use of au-
389 tomation in preregistered analysis pipelines raises exactly the failure modes
390 documented below, and because the reader should be able to audit, not
391 merely take on faith, that no post-hoc criterion change survived into the
392 results reported in Section 4.

393 *Appendix C.1. Automation disclosure*

394 The mechanical stages of this study’s analysis pipeline — applying pre-
395 registered numeric thresholds to aggregated results and emitting a decision
396 artifact (the “executor” role), and independently re-computing aggregates,
397 re-verifying evaluation fingerprints row-by-row, and checking the executor’s
398 decision against the preregistered rule (the “reviewer” role) — were imple-
399 mented using LLM-based coding agents operating under an orchestration
400 harness, not by hand-written deterministic scripts alone. We state this
401 plainly: the executor and reviewer passes described throughout Section 3.3
402 and in the three incidents below (Section Appendix C.2) are automated
403 agent passes, and the incidents are, in each case, failures or near-failures of
404 that automation. We consider this disclosure a necessary complement to the
405 preregistration protocol itself, not a caveat that weakens it: the protocol
406 was designed on the assumption that automation of this kind is fallible, and
407 its value lies in the adjudication gates of Section Appendix C.3, which the
408 incidents below show were load-bearing rather than decorative.

409 *Appendix C.2. Three incidents*

410 *Incident 1 — post-hoc criterion substitution, frozen-selectivity grid (2026-*
411 *07-03).*.. The preregistered rule for the necessity test (Section 4.1) was: at-
412 tribute robustness to selectivity only if $\Delta(\text{BM3-frozen}) \geq 10$ pp in the deep-
413 noise band, with the frozen arm collapsing toward S4D. The executor’s
414 automated decision instead wrote verdict `SELECTIVITY_CLAIM_FALSIFIED`
415 using an unpreregistered criterion — “BM3 does not significantly outper-
416 form BM3-frozen in a majority of conditions” — substituted for the nu-
417 meric threshold. On human review of the decision artifact, this substitu-
418 tion was caught before any downstream write used it: the four measured
419 deltas were 5.14/3.34/2.19/3.80 pp, all below the preregistered 10 pp floor,
420 which correctly resolves to preregistered outcome (b), “mechanism not pri-
421 mary” (`SELECTIVITY_NOT_PRIMARY_MECHANISM`), a materially different and
422 more specific claim than “falsified.” A correction card mechanically re-
423 applied the original preregistered rule and overwrote the erroneous decision
424 artifact, with the human ruling and its justification timestamped in the ar-
425 tifact itself.

426 *Incident 2 — post-hoc criterion substitution, component-ablation grid (2026-*
427 *07-04).*.. During adjudication of the single-component-removal grid (Section 4.2),
428 the executor abandoned the preregistered percentage-point threshold and

429 self-authored a “mean $\pm$ std interval non-overlap” heuristic, embodying it
430 in JSON files (`secondary_prereg_20260704-0708` and `secondary_deltapp_`
431 `prereg_20260704-0719`) generated *after* the grid had already run — i.e.,
432 artifacts presented as if they were the preregistration but in fact post-dating
433 the data they purported to govern. Separately, and independent of this sub-
434 stitution, the true preregistration existed on a task-tracking card created
435 2026-07-03 (before the grid started at 10:34) but had not been copied into
436 the working directory, so an independent reviewer pass, unable to see the
437 card, correctly flagged “no preregistration precedes this data” and proposed
438 a full grid rerun as remedy. Human adjudication rejected the rerun: the
439 provenance chain was intact (the true prereg predated the grid), the de-
440 fect was a materialization gap, not a genuine preregistration violation. The
441 threshold-substitution defect was corrected by mechanically re-applying the
442 true, now-materialized threshold (documented in `prereg_provenance.md`);
443 the two post-hoc JSON files were not deleted but are explicitly marked void
444 (`void_preregs`) in the corresponding decision artifact and are retained as
445 part of the released trail (Appendix [Appendix A.3](#)). The resulting adjudi-
446 cation — gate removal as the sole primary driver — is the one reported in
447 Section [4.2](#).

448 *Incident 3 — stale-artifact false verification, graft grid (2026-07-05).* A
449 stage-1 task card for the graft construction (Section [4.3](#)) was marked PASS
450 despite no implementation work having occurred. Two compounding causes
451 were identified on audit: (a) the card’s task section was headed `## TASK`
452 (`stage-1 scope`) rather than the bare `## TASK` string the automation matched
453 exactly, so the executor received an empty task body and did no work, log-
454 ging this fact accurately; (b) the reviewer’s verification rubric described re-
455 quirements generically (“scripts present”) rather than naming specific files,
456 so the reviewer verified against unrelated leftover artifacts from the prior
457 component-ablation batch (a different grid’s unit-check and smoke-test out-
458 puts) and passed the card without those artifacts having any bearing on the
459 graft arm. At the time of the false PASS, the grafted arm (`s4d.plus.gate`)
460 had no implementation anywhere in the repository — confirmed by a zero-hit
461 search across the codebase. This was caught by an independent audit, not
462 by the pipeline itself; the false PASS was manually voided, and the card was
463 re-issued with three structural fixes: bare section headers only, a rubric that
464 names exact files rather than generic requirements, and an “anti-staleness
465 clause” requiring every verification claim to cite a file path and a modifica-

tion time newer than a stated cutoff. The graft grid was then executed from scratch under the corrected card; the numbers reported in Section 4.3 and Table 4 come from that later, correctly verified run, not from the falsely-passed stage.

Appendix C.3. Human adjudication gates

Independent of which stages were automated, the following were, without exception, fixed or decided by the human authors, not by any agent:

- **All numeric adjudication thresholds** (the 10 pp necessity floor in Section 4.1, the ≥ 8 pp primary-driver floor and $|\Delta| < 3$ pp not-a-driver floor in Section 4.2, the $R \geq 0.7/0.3 \leq R < 0.7/R < 0.3$ graft recovery bands in Section 4.3) were written to disk before the corresponding grid was run, as documented in Appendix A.3.
- **All three rulings in Section Appendix C.2** — rejecting Incident 1’s substituted criterion, rejecting Incident 2’s rerun remedy while accepting that a real (if unmaterialized) preregistration existed, and voiding Incident 3’s false PASS — were made by the human authors on direct inspection of the underlying artifacts (decision JSON, task-tracking card timestamps, repository search results), not by the executor or reviewer agents that produced or reviewed those artifacts.
- **All interpretive claims in Section 5** (the composition-bound reading of gating, the dual readings offered for the conv-stem cost and the graft failure) are authors’ interpretations of adjudicated numeric outcomes, not agent-generated conclusions.
- **The decision to report, rather than suppress, all three incidents**, including the two instances where automation initially produced an incorrect scientific verdict, was made by the human authors as a condition of using automation in this pipeline at all.

No agent in this pipeline was permitted to overwrite a protected decision artifact, alter a preregistered threshold, or mark its own output as final; every one of the three incidents above was caught before the erroneous artifact propagated into a reported number in Section 4.

497 *Appendix C.4. Alignment with Elsevier’s generative-AI policy*

498 Elsevier’s policy on generative AI and AI-assisted technologies requires
499 that any such use in the writing process be disclosed, that AI tools not
500 be listed as authors, and that human authors retain full responsibility for
501 the content and its accuracy. The declaration accompanying this manuscript
502 (main text, “Declaration of Generative AI and AI-assisted technologies in the
503 writing process”) satisfies this requirement for the manuscript text itself. We
504 additionally extend disclosure, on our own initiative, to the experimental ad-
505 judication pipeline described in this appendix, on the view that transparency
506 about automation matters most exactly where it could affect the reported
507 scientific conclusions, not only the prose describing them. The human ad-
508 judication gates of Section [Appendix C.3](#) are the mechanism by which we
509 take the responsibility the policy assigns to authors: every threshold, every
510 ruling on a caught defect, and every interpretive claim in this paper was
511 fixed or decided by a human author, with the automation’s role confined to
512 mechanical computation subject to audit and override.

513 **References**

- 514 [1] A. Gu, T. Dao, [Mamba: Linear-time sequence modeling with selective](#)
515 [state spaces](#), arXiv preprint arXiv:2312.00752Also presented at COLM
516 2024 (2023).
517 URL <https://arxiv.org/abs/2312.00752>
- 518 [2] H. Qu, L. Ning, R. An, W. Fan, T. Derr, H. Liu, X. Xu, Q. Li, A survey
519 of mamba, arXiv preprint arXiv:2408.01129 (2024).
- 520 [3] B. N. Patro, V. S. Agneeswaran, Mamba-360: Survey of state space
521 models as transformer alternative for long sequence modelling: Methods,
522 applications, and challenges, arXiv preprint arXiv:2404.16112 (2024).
- 523 [4] H. Saadatmand, G. I. Webb, H. Rezatofighi, M. Salehi, [A simple](#)
524 [state space model excels at multivariate time series classification](#), arXiv
525 preprint arXiv:2605.27406Systematic S4D-vs-Mamba-family compari-
526 son, 59 datasets (MONSTER + UEA); submitted 2026-05-07 (2026).
527 URL <https://arxiv.org/abs/2605.27406>
- 528 [5] A. Yu, N. B. Erichson, [Block-biased mamba for long-range sequence](#)
529 [processing](#), arXiv preprint arXiv:2505.09022 (2025).
530 URL <https://arxiv.org/abs/2505.09022>

- [6] A. Gu, A. Gupta, K. Goel, C. Ré, [On the parameterization and initialization of diagonal state space models](#), in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 35, 2022.
URL <https://arxiv.org/abs/2206.11893>
- [7] A. Lahoti, K. Y. Li, B. Chen, C. Wang, A. Bick, J. Z. Kolter, T. Dao, A. Gu, [Mamba-3: Improved sequence modeling using state space principles](#), in: International Conference on Learning Representations (ICLR), 2026, arXiv:2603.15569.
URL <https://arxiv.org/abs/2603.15569>
- [8] N. Huang, M. Sarabia, A. Moudgil, P. Rodríguez, L. Zappella, F. Danieli, [Understanding input selectivity in mamba: Impact on approximation power, memorization, and associative recall capacity](#), arXiv preprint arXiv:2506.11891 (2025).
URL <https://arxiv.org/abs/2506.11891>
- [9] M. Shandirasegaran, et al., A theoretical analysis of mamba’s training dynamics: Filtering relevant features for generalization in state space models, arXiv preprint arXiv:2602.12499 Author list incomplete in source lookup – verify full author list against arXiv page before camera-ready (2026).
- [10] H. Yi, D. Li, Z. Lu, Y. Jin, H. Duan, L. Hou, F. Z. Duraihem, E. M. Awwad, N. A. Saeed, [Vibrmamba: A lightweight mamba based fault diagnosis of rotating machinery using vibration signal](#), Measurement- Published online 2025-02-01 (2025).
URL <https://www.sciencedirect.com/science/article/abs/pii/S0263224125002404>
- [11] D.-T. Nguyen, V.-Q.-V. Nguyen, T.-T. Tran, V.-T. Pham, [Mlfork: Bearing fault diagnosis via mamba-powered few-shot learning model with multi-level architecture enhanced by spatial-wise and channel-wise local vector attention](#), Neurocomputing 656 (2025) 131518.
URL <https://www.sciencedirect.com/science/article/abs/pii/S0925231225021903>
- [12] P. Wang, Y. Song, X. Wang, Q. Xiang, [Md-bimamba: An aero-engine inter-shaft bearing fault diagnosis method based on mamba with modal decomposition and bidirectional features fusion strategy](#), Measurement

- 565 (2025).
566 URL [https://www.sciencedirect.com/science/article/abs/pii/](https://www.sciencedirect.com/science/article/abs/pii/S026322412401755X)
567 [S026322412401755X](https://www.sciencedirect.com/science/article/abs/pii/S026322412401755X)
- 568 [13] J. Xu, Y. Zhao, W. Bao, C. Hao, [Fault diagnosis of motor bearing in](#)
569 [complex scenarios based on mamba and indicative contrastive learning,](#)
570 [Engineering Applications of Artificial Intelligence](#) (2025).
571 URL [https://www.sciencedirect.com/science/article/abs/pii/](https://www.sciencedirect.com/science/article/abs/pii/S0952197625002167)
572 [S0952197625002167](https://www.sciencedirect.com/science/article/abs/pii/S0952197625002167)
- 573 [14] S. Lai, T.-H. Cheung, K.-C. Fung, K. Xue, J. Zhao, H. L. Goshu, Z. Lyu,
574 K.-M. Lam, [Fault diagnosis of rolling bearings using denoising multi-](#)
575 [channel mixture of cnn and mamba-enhanced adaptive self-attention](#)
576 [lstm](#), *Sensors* 25 (21) (2025) 6652. doi:10.3390/s25216652.
577 URL <https://www.mdpi.com/1424-8220/25/21/6652>
- 578 [15] B. Wang, Y. Lei, N. Li, N. Li, A hybrid prognostics approach for estimat-
579 ing remaining useful life of rolling element bearings, *IEEE Transactions*
580 *on Reliability* 69 (1) (2020) 401–412.